

Himalaya Singh

M.Tech in Mechanical Design

Phone: 9618142923

Email: aayusingh05@gmail.com

LinkedIn: [linkedin.com/in/himalaya-singh-b29a07236](https://www.linkedin.com/in/himalaya-singh-b29a07236)

Portfolio: <https://me22m817.github.io/himalayasingh.github.io>



Professional Summary and Core Competencies

- Mechanical Simulation Engineer (**M.Tech**) with **2 years** of experience in advanced computational engineering, virtual prototyping, and physics-based product/process development. Experienced in developing robust numerical models for structural response, coupled physical phenomena, and simulation-driven engineering studies to support design validation, performance improvement, and research-oriented analysis.

Technical Skills

- **Cloud & HPC Infrastructure:** Remote visualization (ParaView pvserver), SSH tunneling, client-server architecture, **Docker containerization** ○ **Parallel Computing:** HPC workflows, cloud-bursting
- **Simulation & Modelling:** ANSYS, FEniCS, SfePy ○ **FEM Expertise:** Linear & non-linear structural analysis, thermal analysis, 1-way & 2-way multiphysics coupling, stress analysis
- **Programming & Automation:** Python (NumPy, SciPy, PyVista), Bash scripting for automated simulation pipelines ○ **Operating Systems:** Linux, Windows
- **Geometry & Meshing:** Gmsh, ANSYS Meshing, distributed meshing for large-scale models ○ **Post-Processing:** ParaView, PyVista
- **Design & CAD:** Creo, Fusion 360, Solid Edge, AutoCAD ○ **Regulatory Standards:** ASME, ISO/ASTM, Eurocode

Education

Program	Institution/Board	%/CGPA	Year
M.Tech in Mechanical Design	Indian Institute of Technology Madras	8.06	2022-2024
B.Tech in Mechanical Engg.	GIET (JNTUK)	8.73	2017-2021
HSEB	Mithila Institute of Technology	71.80%	2014-2016
SLC	New Modern Higher Secondary School	76.25%	2014

Experience

Aizant Drug Research Solutions (Smart Development)

June 2024 – Present

Smart Development

Research and Development

Project 1: Compression Testing of Compacted Solids Using Finite Element Analysis

- Established a high fidelity FEM framework combining Drucker Prager plasticity and phase field fracture to capture tablet deformation, crack initiation, and force hardness evolution in close agreement with experimental data.

Project 2: Mathematical Modelling of (FBG) Process to Enhance Machine Performance

- Developed a two-phase (spray and drying) mass- energy balance model incorporating temperature-dependent sorption isotherms and convective heat mass transfer to predict bed temperature and moisture (LOD) evolution for process optimization.

Project 3: 3D Mixing Tank CFD Using OpenFOAM (AMI-Based Rotating Impeller)

- Built and executed a transient incompressible 3D mixing-tank CFD model in OpenFOAM using dynamic mesh motion with AMI for rotating stationary interface coupling, and post-processed velocity, pressure, and temperature fields to evaluate mixing performance and flow regime characteristics.

Key Projects

1. Role of Microstructure on Stress-Diffusion Interactions in Li-Ion Battery Electrodes

Aug 2023 - 2024

M.Tech

IIT Madras

- Simulated stress diffusion coupling during lithiation and delithiation using multiphysics FEM, and analyzed the effects of grain size, boundary thickness, and orientation on stress evolution and failure mechanisms.

2. Design of Worm Gearbox Crane Main Hoist for Uniform Shock Loads.

Jan-April 2023

M.Tech

CAD/CAM Lab

- Designed and modelled a worm gearbox in Creo and Fusion 360, and performed structural and thermal analysis in ANSYS to optimize load handling, durability, and performance.

3. Design and Analysis of Clutch of Hayabusa model 1340.

Jan-April 2023

M.Tech

Design of Mechanical Transmission System

- Designed clutch components in Creo and performed FEA in ANSYS to evaluate stress, thermal effects, and deformation, improving durability and power transmission efficiency.

4. Design & Analysis of heat transfer rate from gas turbine blade using film cooling.

Jan-May 2021

B.Tech

GIET

- Modelled a 3D turbine blade with internal cooling channels in Solid Edge and performed CFD and thermal simulations in ANSYS to optimize cooling performance and blade durability.

5. Fabrication of Forearm machine with electrical power generation.

June-Dec 2020

B.Tech

GIET

- Designed and developed an electromagnetic energy harvesting forearm device, optimizing mechanical design and performance through modelling and simulation using Creo and ANSYS.

Certifications

- Finite element analysis convergence and mesh independence (May, 2023)
- FEM linear, nonlinear analysis and post-processing (May, 2023)
- Automation in robotics control (April, 2021).
- QCAD (Aug, 2020), Solid Edge (Feb, 2020), Python Programming (May, 2023).

Internship

1. Mahindra and Mahindra.

May, 2019

- Hands on exposure to tractor component manufacturing and assembly, including quality control, workflow optimization, and design for manufacturing practices.

2. Design Space System.

June, 2021

- Assisted in CAD modelling, FEA, and CFD simulations to support product design, manufacturability, and performance validation.

Publication

1. Design & Analysis of heat transfer rate from gas turbine blade using film cooling.

July, 2021

IJMTST Journal

Workshops

1. FEniCS (Python Library for Finite Element Methods)

July, 2023

Vanderbilt University

IIT Madras

- Received comprehensive training in the FEniCS FEM library and applied in Li-ion battery and powder compression projects.

2. Computational Fluid Mechanics "Airflow Around a Spoiler".

June-July 2023

Softwares: FEA, SimScale

Online Guided Projects

- Conducted CFD simulations on aerodynamic airflow around an automotive spoiler using SimScale, performing meshing, solver setup, and post-processing of velocity and pressure fields.
- Utilized Sim Scale platform for meshing, solver setup, and post-processing of velocity and pressure fields.

3. CFD Simulation Through a Centrifugal Pump.

June-July 2023

Softwares: FEA, Simscale, msh, CFD.

Online Guided Projects

- Conducted computational fluid dynamics (CFD) analysis to study flow characteristics within a centrifugal pump.

Achievements & Awards

- Certifications of Appreciation (National level Quiz Competitions, May–July 2020):** Basics of Mechanical Engineering (SCET), Strength of Materials (PVPSIT), Engineering Drawing (VIET), Industrial Robotics (NRIIT), Probability & Statistics (VITS)
- Won 1st Prize at MEDHA-19 (GIET, Sept 2019) for designing a portable, backpack mounted agricultural water sprayer system.), MEDHA-19 (GIET) (Sept 2019).**